

Sensors

Sensor: Provides a signal (electrical, optical) depending on the property (physical, chemical, biological) to be measured.

Types:

- Magnetic field sensors
- Distance and position sensors
- Acceleration sensors
- Temperature sensors
- Light sensors
- Pressure sensors
- Force sensors
- Elongation sensors
- Humidity sensors
- Chemical, biological sensors

Features: sensitivity, measuring range, selectivity, non-reactivity, stability.

All around us: Smart devices, Internet of Things.



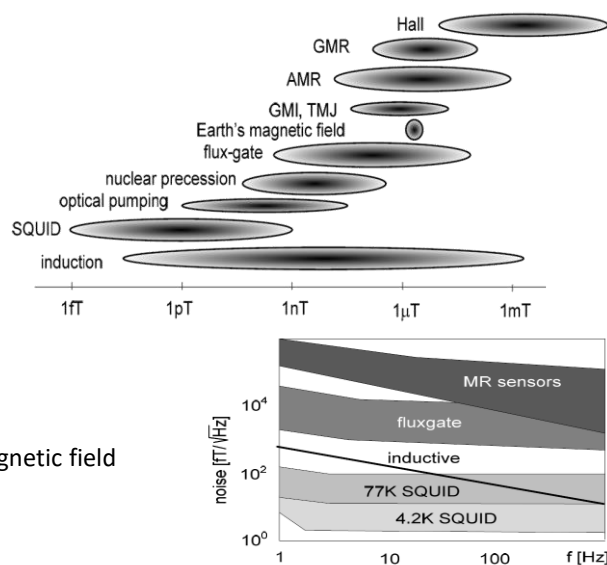
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Magnetic field sensors

- Inductive sensors
- Magnetoresistive sensors: AMR, GMR, TMR
- Hall sensors
- SQUID
- Nitrogen vacancy centers

Typical magnetic field ranges:

- 100fT – 1 pT: human brain's magnetic field
- 30-60uT geomagnetic field
- 10mT: field of a fridge magnet
- 1T: magnetic field in the transformer core
- 1.25T: strong rare earth magnets
- 14T: Field in a superconducting solenoid magnet (e.g. MRI)



Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

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Magnetic field sensors

Factor (tesla)	SI prefix	Value (SI units)	Value (cgs units)	Item
10 ⁻¹⁸	attotesla	5 aT	50 fG	SQUID magnetometers on Gravity Probe B gyroscopes measure fields at this level over several days of averaged measurements ^[2]
10 ⁻¹⁵	femtotesla	2 fT	20 pG	SQUID magnetometers on Gravity Probe B gyros measure fields at this level in about one second
10 ⁻¹²	picotesla	100 fT to 1 pT	1 nG to 10 nG	human brain magnetic field
10 ⁻¹¹		10 pT	100 nG	In September 2006, NASA found "potholes" in the magnetic field in the heliosheath around our solar system that are 10 picoteslas as reported by Voyager ^[3]
10 ⁻⁹	nanotesla	100 pT to 10 nT	1 μG to 100 μG	magnetic field strength in the heliosphere
10 ⁻⁸	microtesla	24 μT	240 mG	strength of magnetic tape near tape head
10 ⁻⁵		31 μT	310 mG	strength of Earth's magnetic field at 0° latitude (on the equator)
		58 μT	580 mG	strength of Earth's magnetic field at 50° latitude
10 ⁻³	millitesla	0.5 mT	5 G	the suggested exposure limit for cardiac pacemakers by American Conference of Governmental Industrial Hygienists (ACGIH)
		5 mT	50 G	the strength of a typical refrigerator magnet ^[4]
10 ⁻¹		0.15 T	1.5 kG	the magnetic field strength of a sunspot
		1 T to 2.4 T	10 kG to 24 kG	coil gap of a typical loudspeaker magnet ^[5]
		1 T to 2 T	10 kG to 20 kG	inside the core of a modern 60 Hz power transformer ^{[6][7]}
10 ⁰	tesla	1.25 T	12.5 kG	strength of a modern neodymium-iron-boron (Nd ₂ Fe ₁₄ B) rare earth magnet. A coin-sized neodymium magnet can lift more than 9 kg, pinch skin and erase credit cards. ^[8]
		1.5 T to 3 T	15 kG to 30 kG	strength of medical magnetic resonance imaging systems in practice, experimentally up to 8 T ^{[9][10]}
		9.4 T	94 kG	Modern high resolution research magnetic resonance imaging system
		11.7 T	117 kG	field strength of a 500 MHz NMR spectrometer
		16 T	160 kG	strength used to levitate a frog ^[11]
10 ¹		23.5 T	235 kG	field strength of a 1 GHz NMR spectrometer ^[12]
		36.2 T	362 kG	strongest continuous magnetic field produced by non-superconductive resistive magnet ^[13]
		45 T	450 kG	strongest continuous magnetic field yet produced in a laboratory (Florida State University's National High Magnetic Field Laboratory in Tallahassee, USA). ^[14]
10 ²		129.053 T	1.29 MG	strongest (pulsed) magnetic field yet obtained non-destructively in a laboratory (Kesho Foundation, SSI, with less than 20 watts of power and 10ml or less of gans nano fluid , ITALY) ^[15]
		730 T	7.3 MG	strongest pulsed magnetic field yet obtained in a laboratory, destroying the used equipment, but not the laboratory itself (Institute for Solid State Physics, Tokyo)
10 ³	kilotesla	2.8 kT	28 MG	strongest (pulsed) magnetic field ever obtained (with explosives) in a laboratory (VNIIEF in Sarov, Russia, 1998) ^[16]

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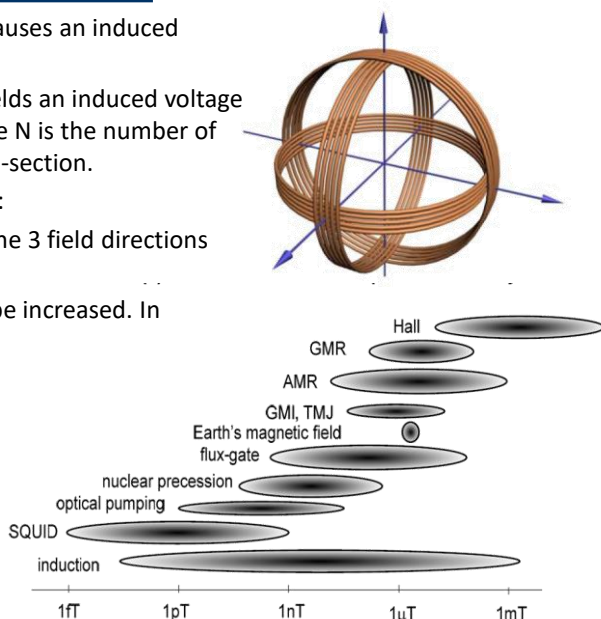
Magnetic field sensors: inductive sensors

Principle: a variable B field in a coil causes an induced voltage.

$B = B_0 \sin \omega t$ oscillating field yields an induced voltage amplitude: $V_0 = -nA\omega B_0$, where N is the number of turns of the coil and A is the cross-section.

Advantages (+) and disadvantages (-):

- + Broad application range, simple, the 3 field directions are easily detectable
- To increase sensitivity, size must be increased. In geophysics meter-sized inductive sensors are also used.
- Detection requires changing magnetic field.
→ to measure static field the sensor must be moved



Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

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Magnetoresistive magnetic field sensors

Magnetic field sensors based on resistance change.

They can rely on several physical phenomena:

- AMR**: Anisotropic Magnetoresistance
- GMR**: Giant Magnetoresistance, **SV**: Spin Valve
- MTJ**: Magnetic Tunnel Junction
- CMR**: Colossal Magnetoresistance

AMR, Anisotropic Magnetoresistance

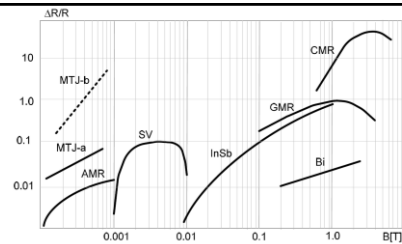
1857 Kelvin: The resistance of a ferromagnet (F) is different if the current direction is parallel or perpendicular to the direction of magnetization.

Reason: It is due to magnetism and so-called spin-orbit interaction. For example, the electron scatters more strongly when it moves in the direction of M . In general, the resistance is higher in the direction of M .

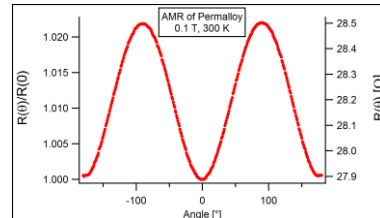
Typical angle dependence of the AMR phenomenon:

$$\rho(\theta) = \rho_{\text{perp}} + (\rho_{\text{paral}} - \rho_{\text{perp}}) \cos^2(\theta),$$

where θ is the angle between the M magnetization and the j current density. The ρ resistivity is ρ_{perp} or ρ_{paral} if the current is perpendicular or parallel to M .



Various physical phenomena associated with magnetic-field-induced resistance change. Relative resistance changes are shown as a function of magnetic field covering various field ranges.



Typical AMR signal: angle dependence of the resistance in Permalloy (20% Fe, 80% Ni)

Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

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Anisotropic Magnetoresistance (AMR) magnetic field sensors

In the case of a thin narrow but long ferromagnetic layer, the shape anisotropy determines the preferred magnetization direction. For example, magnetization sets in the longitudinal direction (see the easy axis in the figure).

Principle: an external field is applied perpendicular to the internal magnetization direction. This field somewhat modifies the magnetization direction, i.e. the angle of the current and M , and the related magnetoresistance changes.

Problem: naturally the current would also flow in the longitudinal direction, yielding $\theta=0$. At this working point ρ shows a quadratic θ dependence (see previous slide) instead of a linear relation. The best linear relation is obtained at $\theta=45^\circ$.

Solution: Barber-pole geometry. We evaporate well conducting metallic stripes on the top of the magnetic layer with 45° orientation. These stripes are much better conductors than the magnetic layer underneath, i.e. the current wishes to minimize its route between two neighbor metallic stripes, i.e. it flows at $\theta=45^\circ$ direction in the magnetic layer in those segments, where no metal is on the top. Otherwise, it flows in the metallic stripe instead of the magnetic layer.

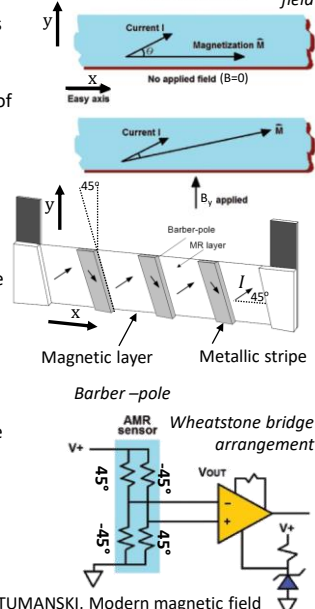
Problem 2: the resistance of the AMR sensor also depends on the temperature.

Practical implementation: Wheatstone – bridge with Barber poles of 45° and -45° orientation. V_+ is applied, and V_{OUT} is measured. In this arrangement, the resistance of the 4 AMR sensors will change similarly with the external temperature change.

Advantages, disadvantages:

- + Simple, long-known principle, cheap.
- + Better sensitivity than GMR sensors. E.g. Phillips KMZ10B AMR sensor has $16\mu\text{V}/\mu\text{T}$ sensitivity at $V_+=5\text{V}$ supply voltage.
- Small resistance change, $<2\%$
- Sensitive to longitudinal field components. Operational condition: $B_y < 0.1B_x$
- In large B field the sensor can be demagnetized $\rightarrow$ malfunction

Magnetization and current direction in a ferromagnetic (F) tape without and in the presence of an external magnetic field



S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

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Giant Magnetoresistance (GMR) magnetic field sensors

Nobel prize 2007: Albert Fert & Peter Grünberg

Arrangement: Ferromagnetic (FM) thin layers separated by nonmagnetic (NM) metallic layer. In $B=0$ field the FM layers align antiferromagnetically (neighbor FM layers have opposite, antiparallel, AP magnetization). In large field all the FM layers align with the field direction (parallel, P magnetization).

Operation: if the magnetization is opposite to the electron spin, the electrons scatter a lot in the F layer, yielding a large resistance ($R_{\uparrow\downarrow} = R_{\downarrow\uparrow}$). The same magnetization and spin orientation yields a smaller resistance due to the less scattering ($R_{\uparrow\uparrow} = R_{\downarrow\downarrow} < R_{\uparrow\downarrow}$). The phenomenon is similar to optical polarizing filters: little light passes through crossed polarizing filters, while much light passes through parallel ones. Here the $\uparrow\downarrow$ spin and magnetization orientation filters the electrons. It can be seen in a simple resistor model, that in large field the resistance is smaller (the structure is more transparent for at least one of the spin orientations, whereas in zero field both spin orientations are filtered in one of the layers).

Source: wikipedia

Spin direction magnetization direction

$$R_{B=0} = R_{AP} = \frac{R_{\uparrow\downarrow} + R_{\downarrow\uparrow}}{2} \quad R_{\text{large } B} = R_P = \frac{2R_{\uparrow\uparrow}R_{\downarrow\downarrow}}{R_{\uparrow\downarrow} + R_{\downarrow\uparrow}}$$

$$R_{\uparrow\uparrow} = R_{\downarrow\downarrow} = \bar{R}(1 - \beta), \quad R_{\uparrow\downarrow} = R_{\downarrow\uparrow} = \bar{R}(1 + \beta)$$

$$R_{B=0} = \bar{R} \quad R_{\text{large } B} = \bar{R}(1 - \beta^2) < R_{B=0}$$

Variation of the resistance and the magnetization orientation with the magnetic field

Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)

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Original GMR, Spin valve and tunneling magnetoresistance sensors

Original GMR arrangement: multiple ferromagnetic layers used with a very thin nonmagnetic layer between them so that an antiferromagnetic coupling is established between the neighbor layers. A large field is required to align the magnetizations, i.e. the sensor is not very sensitive. The ΔR resistance change can reach 50%.

Spin valve (SV) geometry: This is more common nowadays. There is a freely rotating FM layer, and another FM layer with magnetization pinned by a neighbor layer. In this arrangement the NM spacer layers should not be so thin as in the original geometry, as the magnetization alignment is not determined by this. The SV arrangement is used in the read heads of hard disk drives (HDD) and also in magnetic memories (MRAM).

Advantages and disadvantages of the SV geometry:

- + Small, more sensitive than the original GMR geometry, small energy consumption, fast response, weak temperature dependence
- More expensive

Tunneling Magnetoresistance (TMR) or Magnetic Tunnel Junction (MTJ) sensors:

Thin insulator between the two FM layers.

- + $\Delta R/R$ can reach 1000% in MgO tunnel junctions. Sensitive already in the mT field range.
- Expensive heterostructure, small B range, large energy consumption

conduction electrons

Cu Co Cu Co Cu Co Cu

AF (antiferromagnet)
FM - pinned
NM
FM - free

spin valve

sensing current

recorded bits

ferromagnetic electrodes
insulator
external magnetic field (H)

large tunnel current

small tunnel current

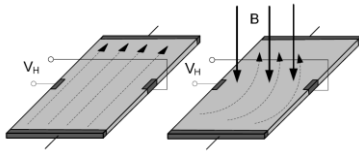
$TMR = \frac{P_1 P_2}{1 - P_1 P_2}$

Sources: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)
<https://www.electricalibrary.com/en/2017/08/02/what-is-spintronics/>

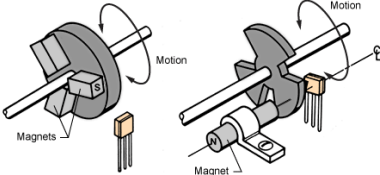
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Hall sensors

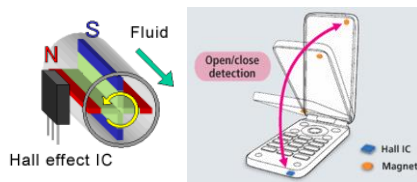
Hall voltage due to the Lorentz force in B field



Hall sensors as revolution counters. Magnets on the axle, or fixed magnet with rotating shielding disc



Flow meter (left) and openness sensor (right)



<http://www.allegromicro.com/en/Design-Center/Technical-Documents/Hall-Effect-Sensor-IC-Publications/Hall-Effect-IC-Application-Guide.aspx>

1879 Hall effect (see the slides on Quantized Hall effect for more details)

Principle: in B field the Lorentz force yields a perpendicular voltage drop on the sample (Hall voltage, V_H). The related Hall resistance in a 2D electron gas (see Metrology slides):

$$R_H = \frac{V_H}{I} = \frac{E_y}{j_x} = \frac{B}{en_{2D}}$$

Small n electron density is required for a large signal $\rightarrow$ semiconducting samples should be used.

Advantages/disadvantages:

- + Simple and cheap, the market is dominated by Hall sensors
- Not too sensitive, $<5\text{mV/mT}$ (but 50mV/mT is also achievable)
- Very strong temperature dependence due to the semiconducting nature of the sample
- + Size can be smaller than $1\mu\text{m}$,
- + Small energy consumption
- + Non-invasive measurement, i.e. no ferromagnetic part which generates some magnetic field. Also lacks the demagnetization problem of the sensors with FM layers.
- + Si-based sensors are easily embedded in integrated circuits with amplifiers, stabilizers

Typical applications:

- Revolution counters e.g. in cars.
- Flow meters
- Openness sensors for mobile phones (very low power consumption $<1.4\mu\text{A}$)

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Comparison of various magnetoresistive sensors

Typical price: $<1\text{USD/piece}$

• Hall sensor

- Small voltage signal and small sensitivity, strong temperature dependence

+ Cheap, small, low consumption

• GMR and SV

- + Small, sensitive, weak temperature dependence, low consumption, fast

- More expensive to produce

- Hard to measure field polarity, not very good for a compass

• TMR

- + Sensitive, small noise

- Narrow B field range, larger energy consumption, expensive

• AMR

2-3% R variation, Wheatstone bridge arrangement, mV output

- + high accuracy, weak temperature dependence

- Small signal, but it can be amplified. Large size.

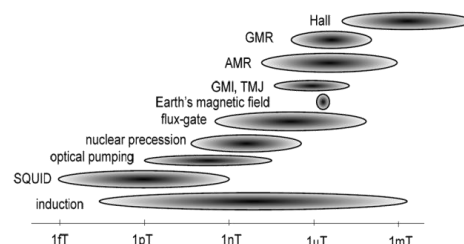
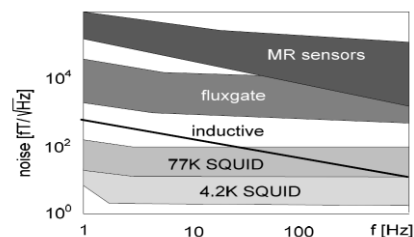
A Further compact, but non-magnetoresistive sensor:

- **Lorentz force meter** (new Microelectromechanical System, MEMS sensor)

Detection of a small magnet's displacements due to magnetic field

- + CMOS compatible, easily integrated

- Resolution is limited by the noise of the electronics



Y. Cai et al. Magnetometer basics for mobile phone applications, Sensors & Transducers

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Superconducting Quantum Interference Device (SQUID)

Principle: two parallel Josephson junctions (JJ). In zero field the phase difference would be the same δ_0 at the two JJs. A finite field introduces a different phase difference at the two arms: $\delta_1 = \delta_0 + \frac{2\pi e}{h} \Phi$, $\delta_2 = \delta_0 - \frac{2\pi e}{h} \Phi$, where e is the electron charge, h is Planck's constant, and $\Phi = BA$ is the enclosed flux with B being the magnetic field and A the area enclosed by the SQUID. Like this the total current of the SQUID is written as:

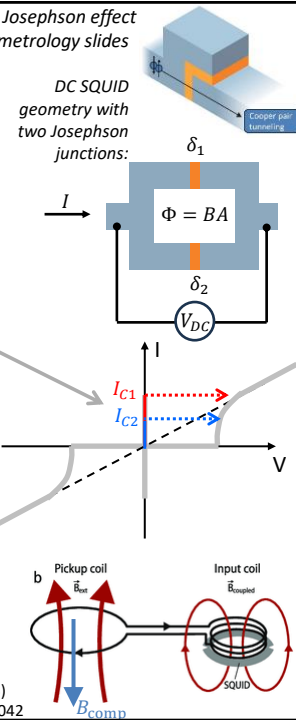
$$I = I_1 + I_2 = I_0 [\sin(\delta_0 + 2\pi e \Phi / h) + \sin(\delta_0 - 2\pi e \Phi / h)] = 2I_0 \cos(2\pi e \Phi / h) \sin \delta_0$$

This means, that the critical current of the SQUID is modulated by the enclosed flux according to the $\cos(2\pi e \Phi / h)$ function.

If we current bias the squid and increase the current, zero voltage drops on the SQUID as long as the critical current is not reached. Reaching the critical current, the SQUID turns to a non-superconducting state, where already a finite voltage drop is observed. As the magnetic field is varied, this crossover happens at different currents, and thus different voltages. This is periodic in the flux: taking $A=1\text{cm}^2$ device area one period corresponds to $\Delta B=2 \cdot 10^{-11}$ T change in the magnetic field. This is an extremely sensitive device for measuring very small, femto-tesla field variations!

Usually, a loop of a closed superconducting wire is applied as a flux transformer. Due to its zero resistance the superconducting wire will preserve the overall enclosed flux (see demo experiment in the first lecture). The flux detected by the SQUID at the input coil equals the flux enclosed by the pickup coil divided by the number of turns at the input coil. Usually a feedback is also used: a B_{comp} compensating field keeps the flux zero at the pickup, and accordingly also at the SQUID. If the SQUID detects field variation, the compensating field is regulated with a PI loop to keep the flux zero.

see the Josephson effect on the metrology slides



Source: S. TUMANSKI, Modern magnetic field sensors – a review PRZEGLĄD ELEKTROTECHNICZNY (2013)
IFCN-endorsed practical guidelines for clinical magnetoencephalography, DOI:10.1016/j.clinph.2018.03.042

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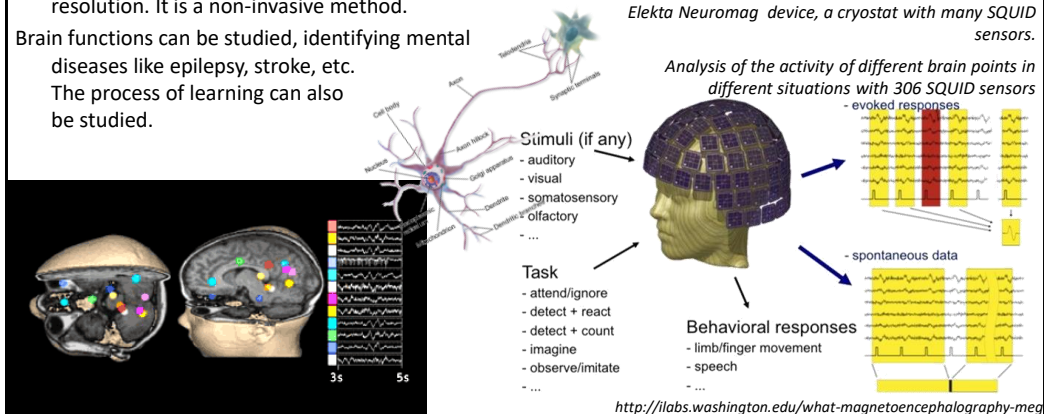
Superconducting Quantum Interference Device (SQUID) - applications

- Extremely sensitive, fT fields can be detected!
- Important application area: investigation of brain functions

MEG (Magnetoencephalography)

Advantages of MEG in brain research: both position and time of brain activity can be tracked in situ with $\sim\text{mm}$ spatial and $\sim\text{sub-ms}$ temporal resolution. It is a non-invasive method.

Brain functions can be studied, identifying mental diseases like epilepsy, stroke, etc. The process of learning can also be studied.



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